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Neuroanatomy: Olfaction and Taste

- Gonzalo Otazu, Ph.D.
- Biomedical Sciences

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

- Session Objectives

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1. Describe the basic cellular organization of the human olfactory and gustatory systems and relate this information to the functional anatomy of each system.

2. Explain how afferent and efferent olfactory and gustatory pathways contribute to the communication of those special senses between sensory epithelia and brain.

3. Correlate the input of specific brain with the complexity of smell/taste sensation.

4. Compare and contrast both systems on a cellular, anatomical and functional level.

5. Apply specific disease states (induced or inherited) to each system and predict their clinical impact.

Olfaction

Clinical Relevance

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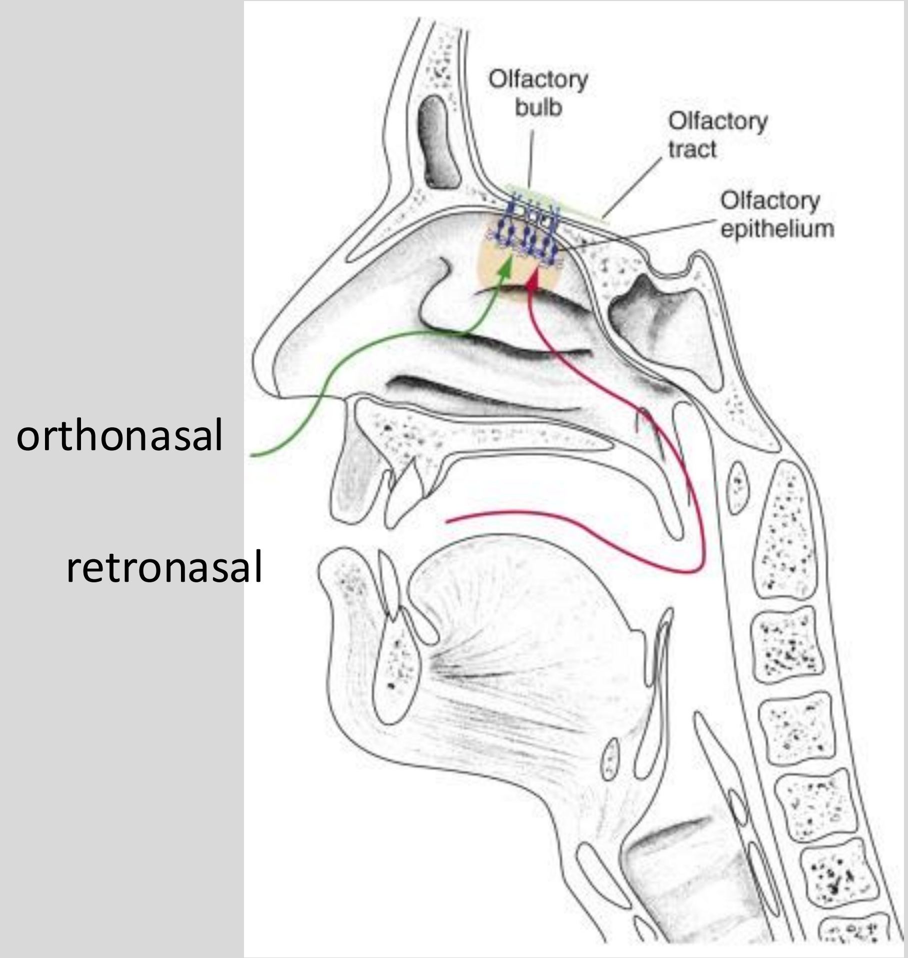
- Parkinson patients show olfactory deficits 4 years before showing motor symptoms.

Source:http://media.neurologyadvisor.com/images/2015/11/17/womanflowers_869683.jpg?format=jpg&zoom=1&quality=70&anchor=middlecenter&width=320&mode=pad

The Human Olfactory System

- *Retronasal olfaction* is the perception of odors emanating from the oral cavity during eating and drinking, as opposed to *orthonasal olfaction*, which occurs during sniffing.
- Retronasal olfaction is important for the flavor sensation= taste+olfaction



Source: Saive et al., Front. Behav. Neurosci., 07 July 2014 | http://dx.doi.org/10.3389/fnbeh.2014.00240

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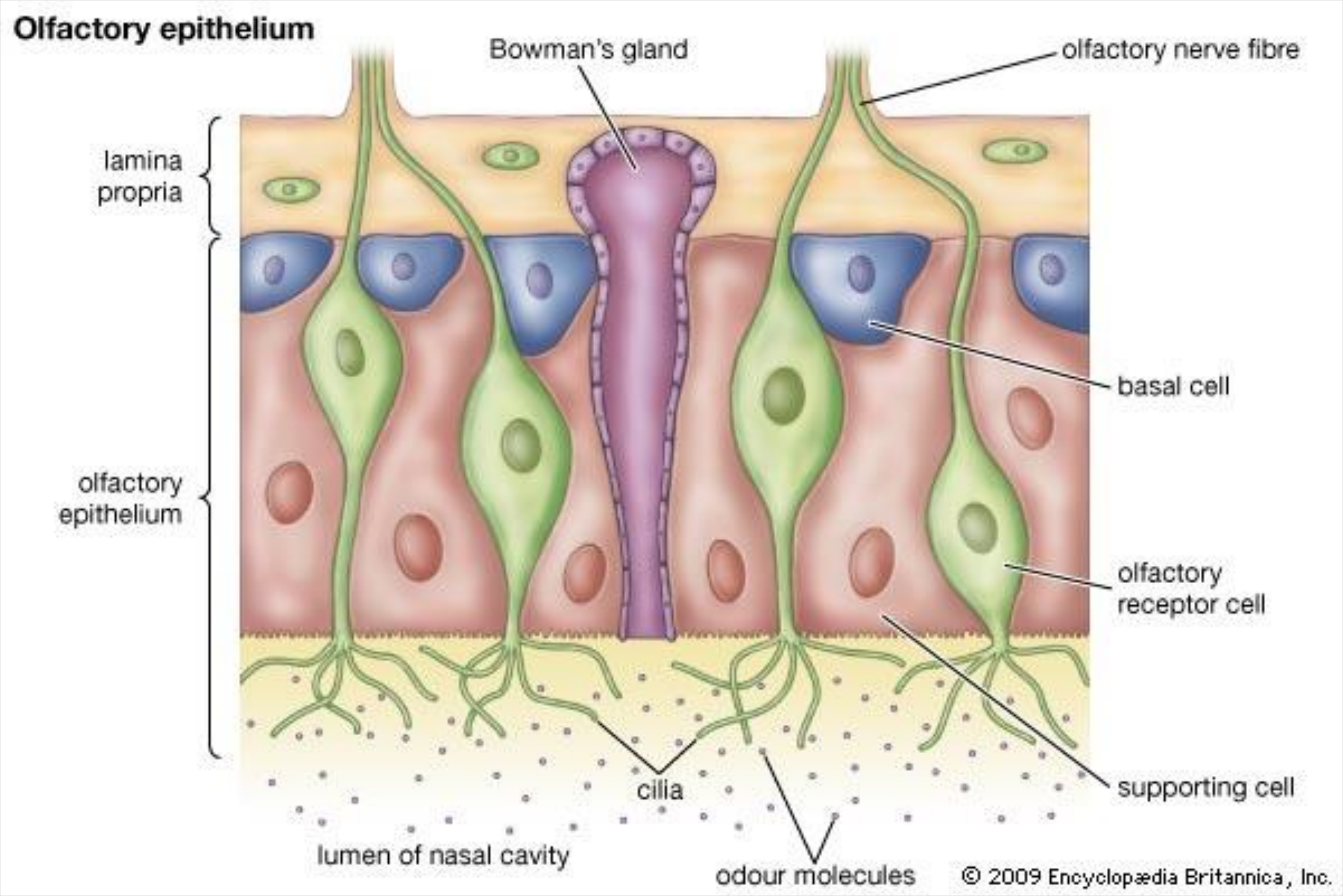
- Olfactory Epithelium - Overview



Source: Encyclopedia Britannica Inc., 2009

Olfactory Cells

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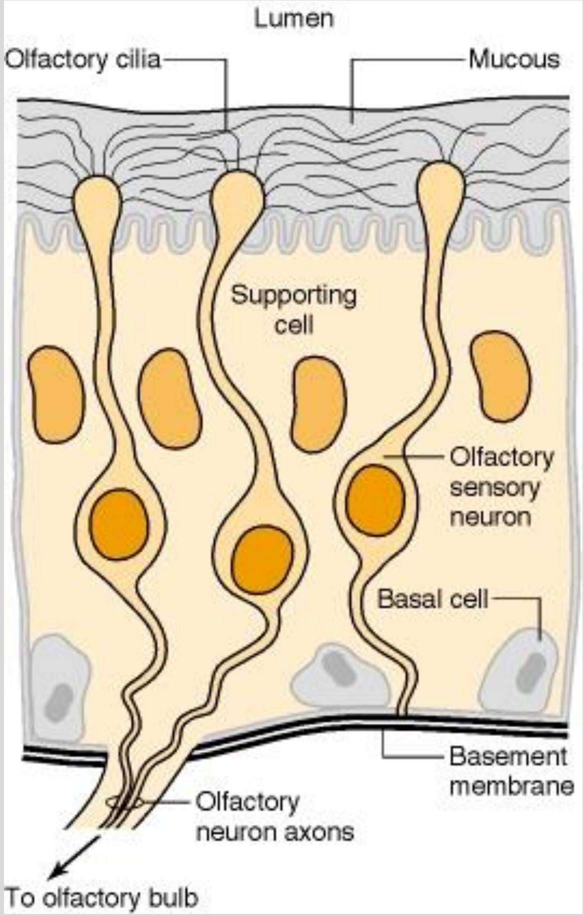
- First order bipolar neurons
- A.K.A. Olfactory neurosensory cells, primary olfactory neurons
- Dendrites with specialized cilia

  Short-lived, high turnover (7 days)
- Chemotherapy affects smell perception

<u>Basal cells:</u>

- Stem cells that give rise to the olfactory receptor cells



Source: Siegel, Essential Neuroscience, 3rd edition

Olfactory Epithelium - Supporting Cells

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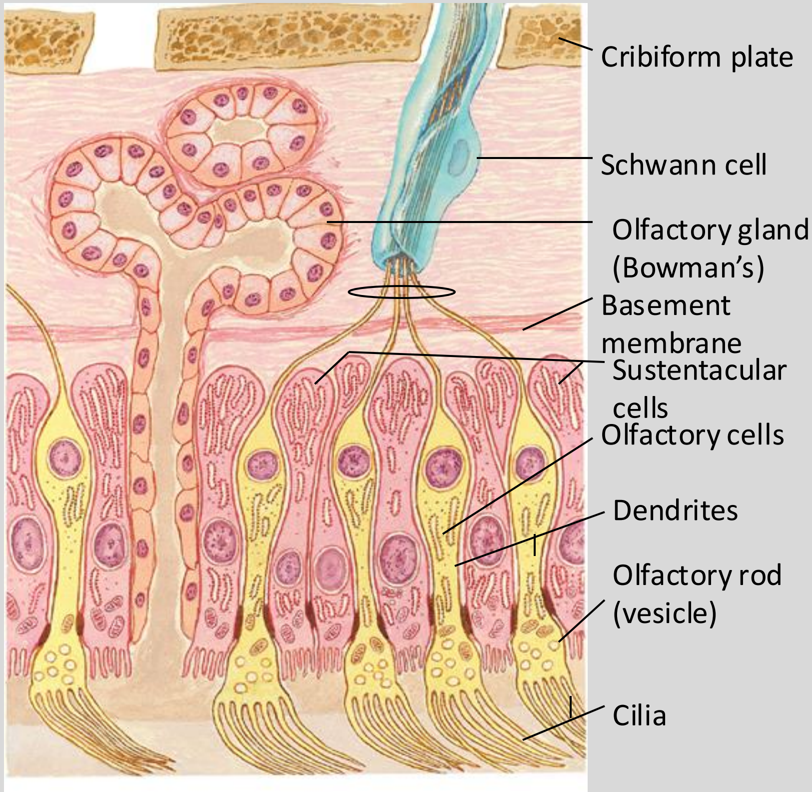
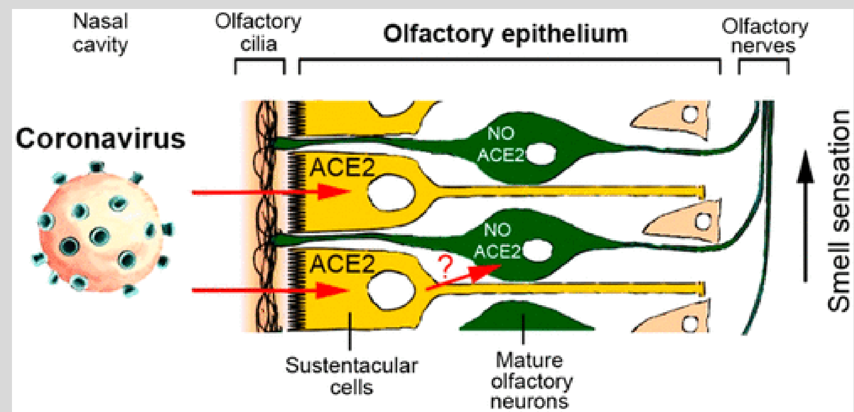


<u>Sustentacular cells:</u>

- Microvilli and secretory granules
- Mucous
- Structural stability

<u>Cells associated with Bowman's glands</u>:

- Secrete mucous fluid containing glycoproteins



Expression of the SARS-CoV-2 Entry Proteins, ACE2 and TMPRSS2, in Cells of the Olfactory Epithelium: Identification of Cell Types and Trends with Age
Katarzyna Bilinska ,Patrycja Jakubowska ,Christopher S. Von Bartheld , and Rafal Butowt*
ACS Chem. Neurosci. 2020,

- Olfactory receptors are G-coupled protein receptors

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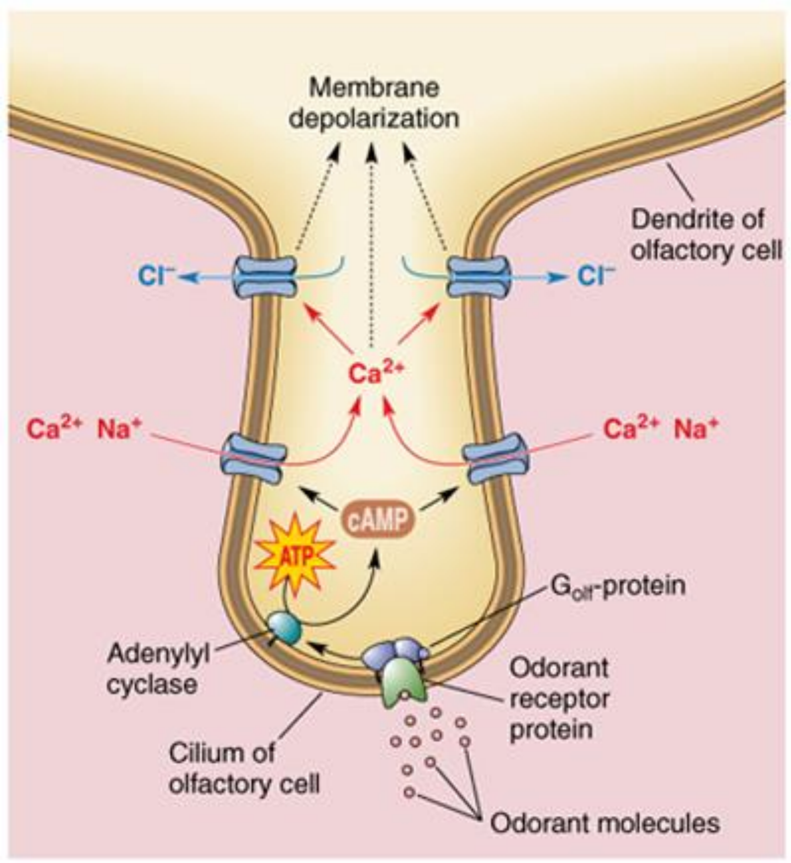

-Olfactory receptor protein binds an array of odorants
-Humans have ~300 olfactory receptor genes (compared to mice ~1200)
.-A receptor-odorant complex is formed, which activates a G protein.
-The second messenger cascade causes the opening of cation channels (Na^{+}, K^{+} and **Ca^{2+}**)
-Ca^{2+} influx causes the opening of Ca^{2+}-gated Cl^{-} channels opening Cl^{-} channels causes depolarization of an olfactory receptor neuron and amplifies the receptor potential. The depolarization spreads to the trigger zone of the receptor neuron and, if large enough, initiates action potentials

Source: Nolte, The Human Brain: An Introduction to its Functional Anatomy, 7th Edition, 2016

Olfactory Nerve: CNI

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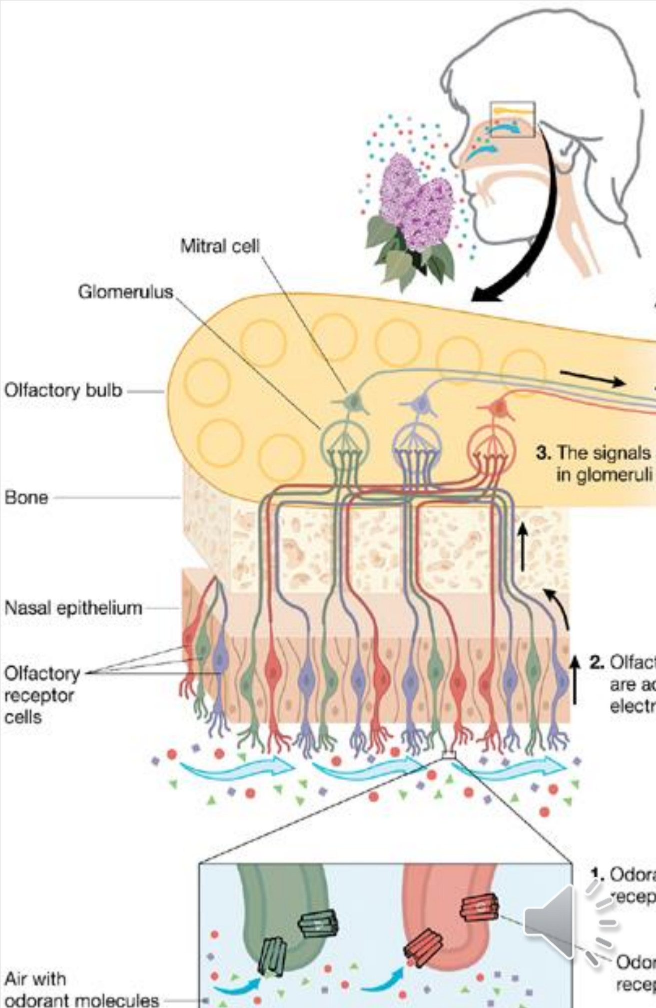
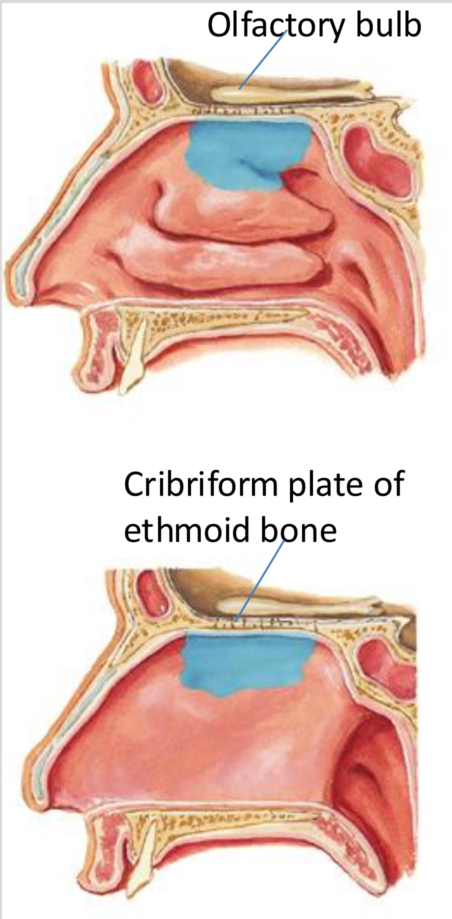
Unmyelinated axons of the olfactory cells gather into about 20 bundles - the olfactory fila

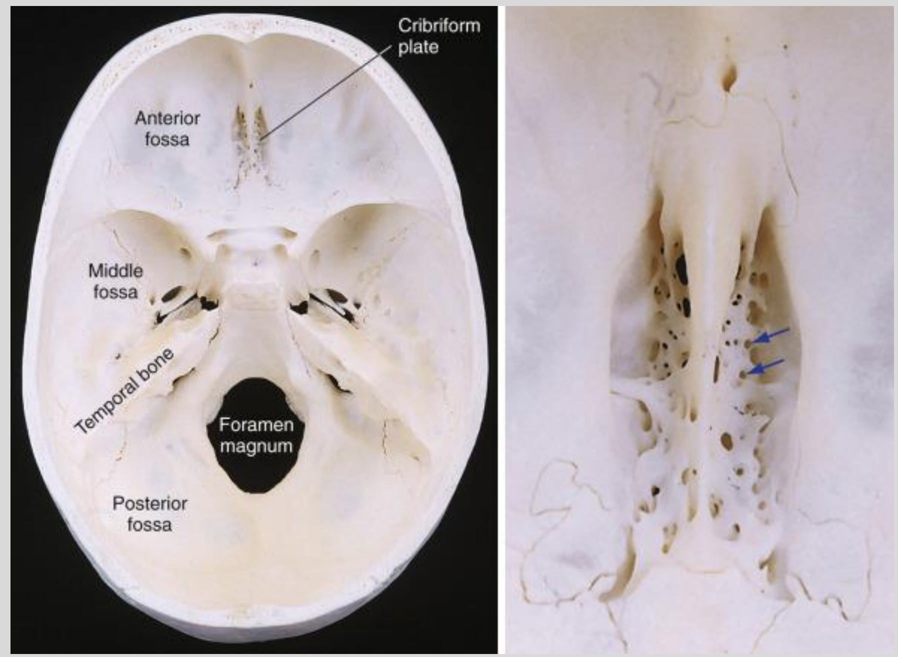
Traverse through foramina in the cribiform plate in the ethmoid bone

Synapse in the olfactory bulb in glomeruli



- Cranial nerve I crosses through the cribriform plate to reach the olfactory bulb





Nolte's The Human Brain, 13, 329-347

- Olfactory receptor neurons expressing the same receptor converge to the same glomerulus in the olfactory bulb

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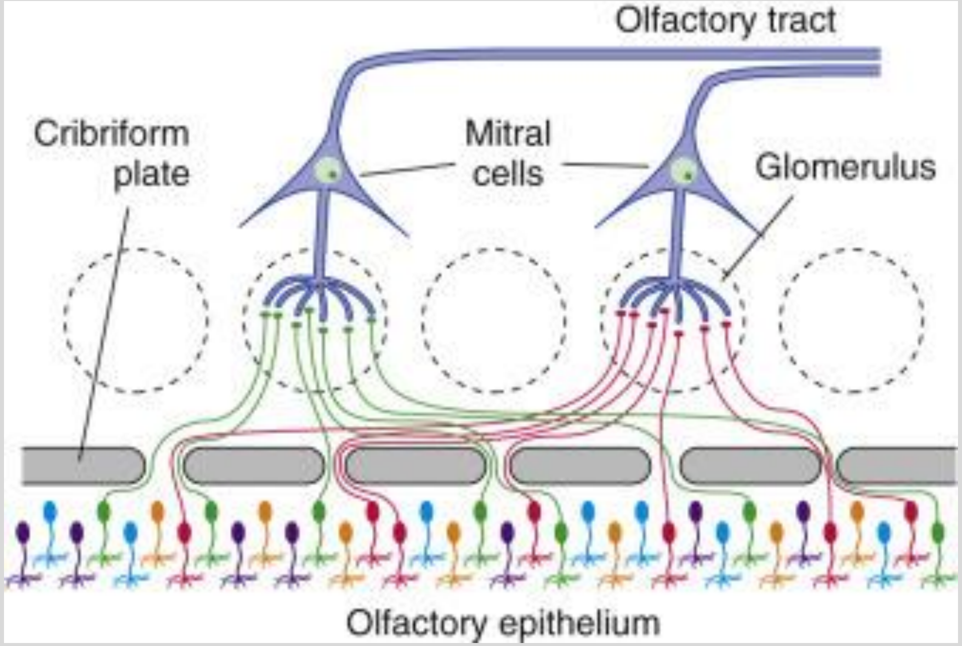
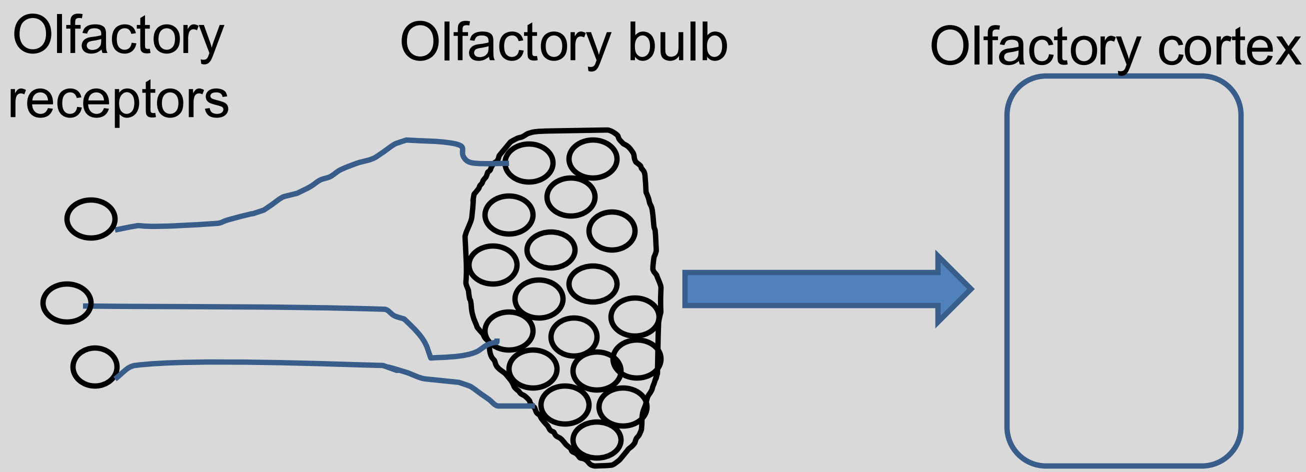
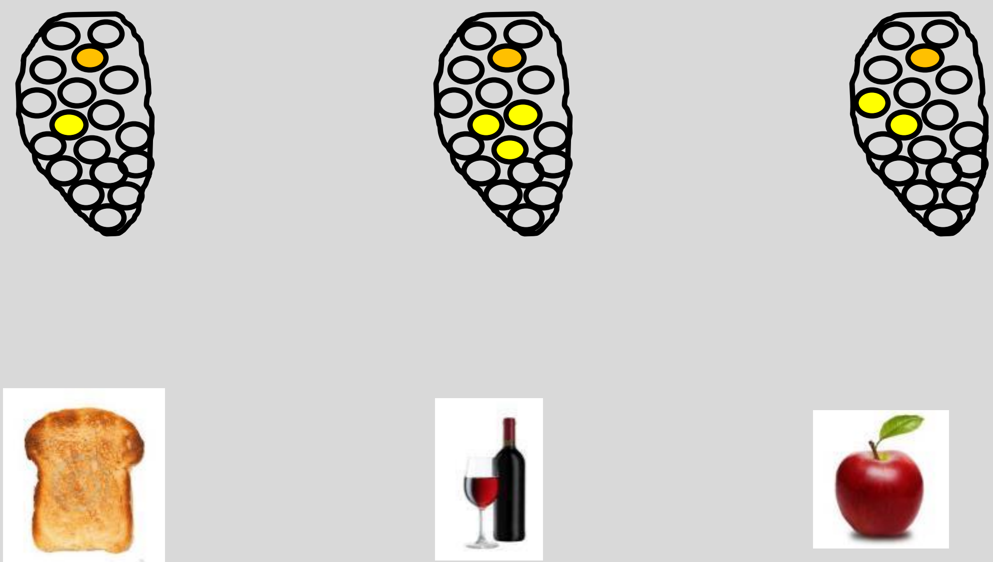


-Olfactory receptor neurons project to the olfactory bulb, forming a structure called a glomerulus.
-Each glomerulus receives input from a single class of olfactory receptor.
-An single odorant might active different types of receptors.

Source: Modified from Cranial Nerves 3rd Ed., 2010 Wilson-Pauwels, Stewart, Akesson, Spacey, PMPH-USA

Odors activate a distinct pattern of olfactory receptors





Glomerular activation in response to ethyl butyrate0.1%

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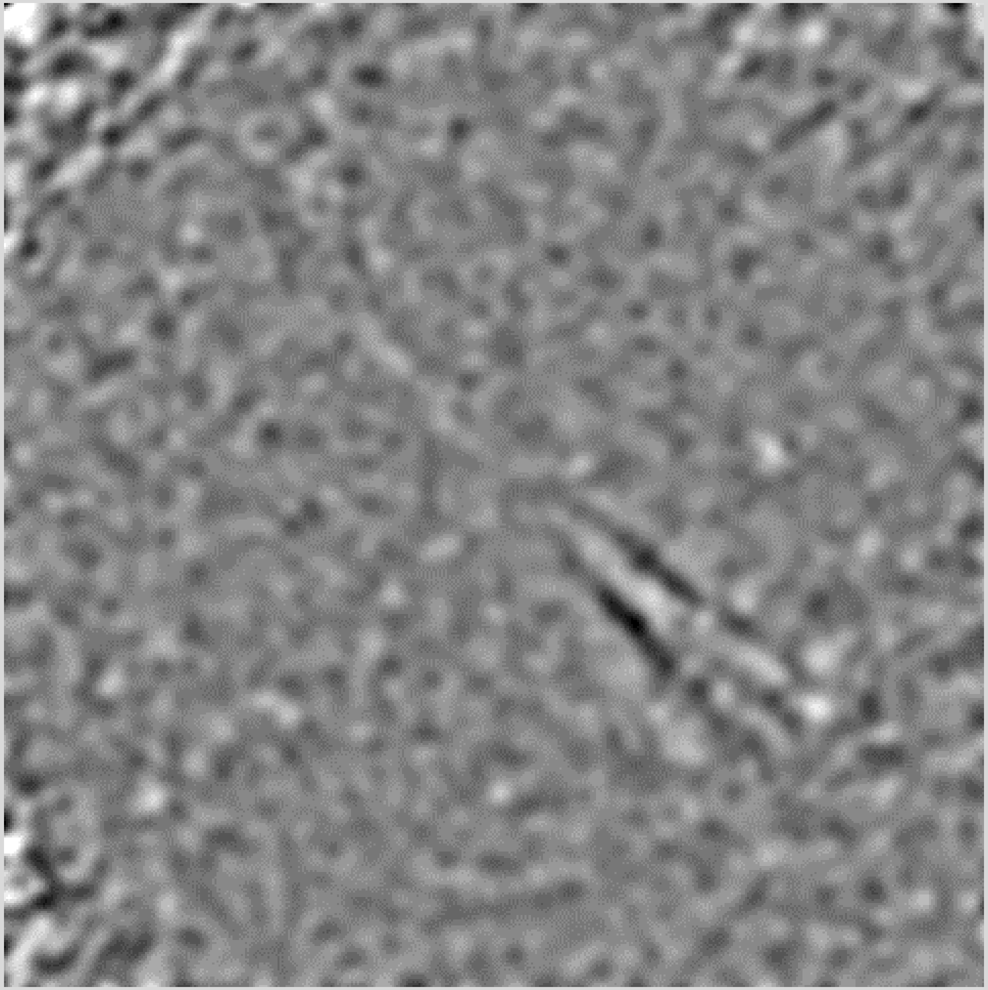
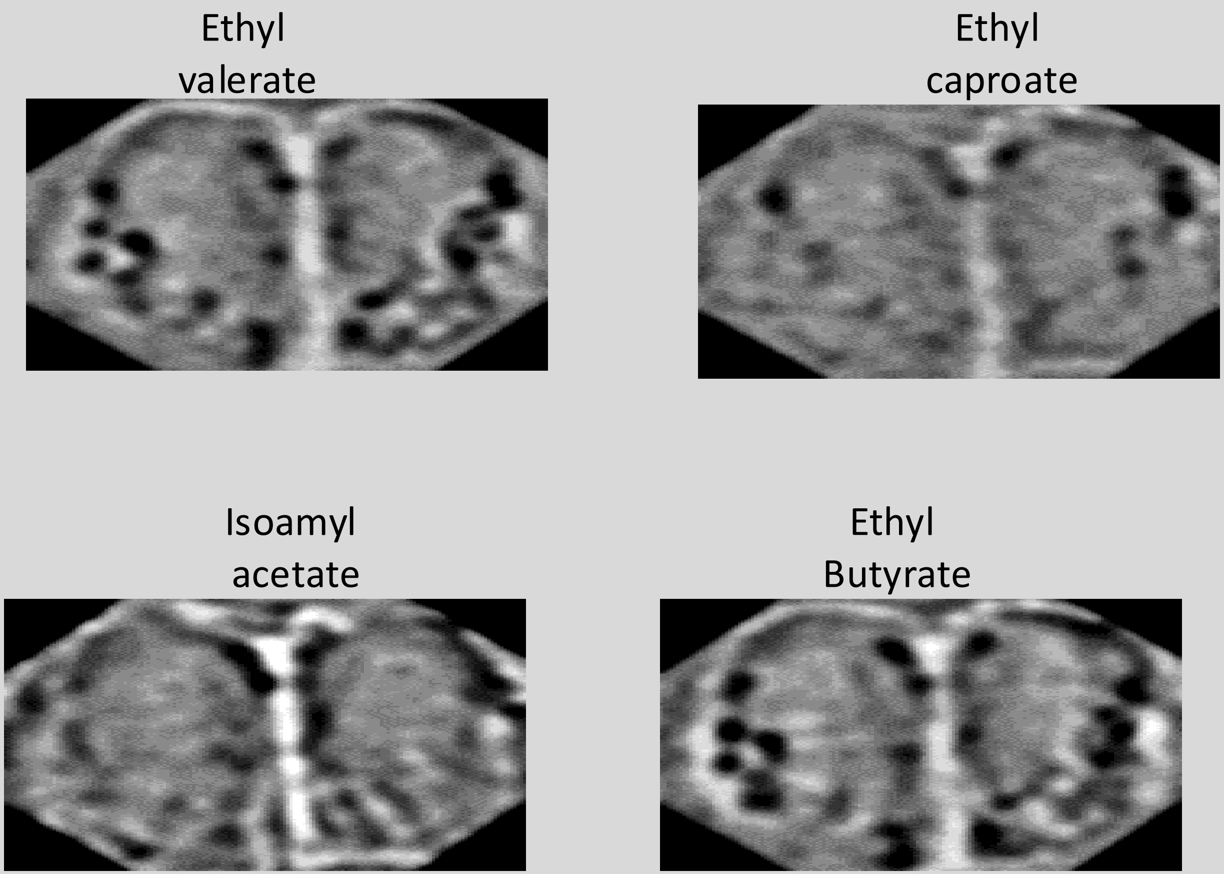


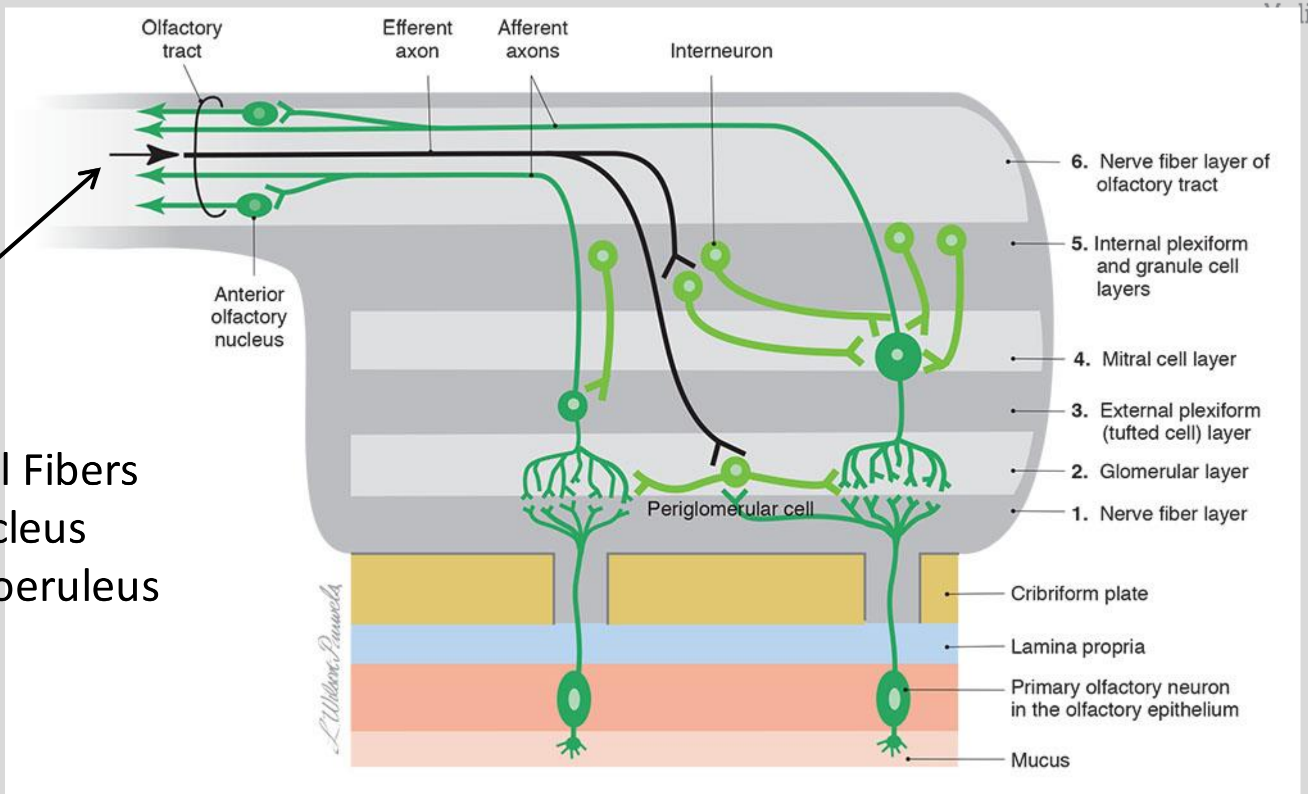
Li et at, Nature Communications, 2023

Odors activate overlapping patterns of olfactory receptors



- Different odors can activate overlapping sets of glomerulus.
- The combination of olfactory glomeruli determines the identity of an odor.

- Olfactory bulb projections



- Centrifugal Fibers
- Raphe Nucleus
- Nucleus Coeruleus

- Glomeruli are contacted by dendrites of mitral and tufted cells.
- These cells are the output of the bulb and transmit the olfactory information to other parts of the brain.
- The activity of mitral and tufted cells are regulated by interneurons of the olfactory bulb: granule cells and the periglomerular cells.
- These interneurons are regulated by descending projections from piriform cortex, anterior olfactory nucleus, and neuromodulatory systems.

Source: Modified from Cranial Nerves 3rd Ed., 2010 Wilson-Pauwels, Stewart, Akesson, Spacey, PMPH-USA

Olfactory Tracts

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- Axons of mitral and tufted cells
- Olfactory tract expands at the Olfactory trigone
- Bifurcates into the Lateral and Medial stria
- Lateral tract has the axons of the principal cells and projects to the Primary olfactory cortex
- Medial olfactory stria project to the contralateral anterior olfactory nucleus.
- Anterior olfactory nucleus is an intermediate nucleus located in the olfactory tract.
- Anterior olfactory nucleus receives information from both bulbs and might play a role in olfactory localization.

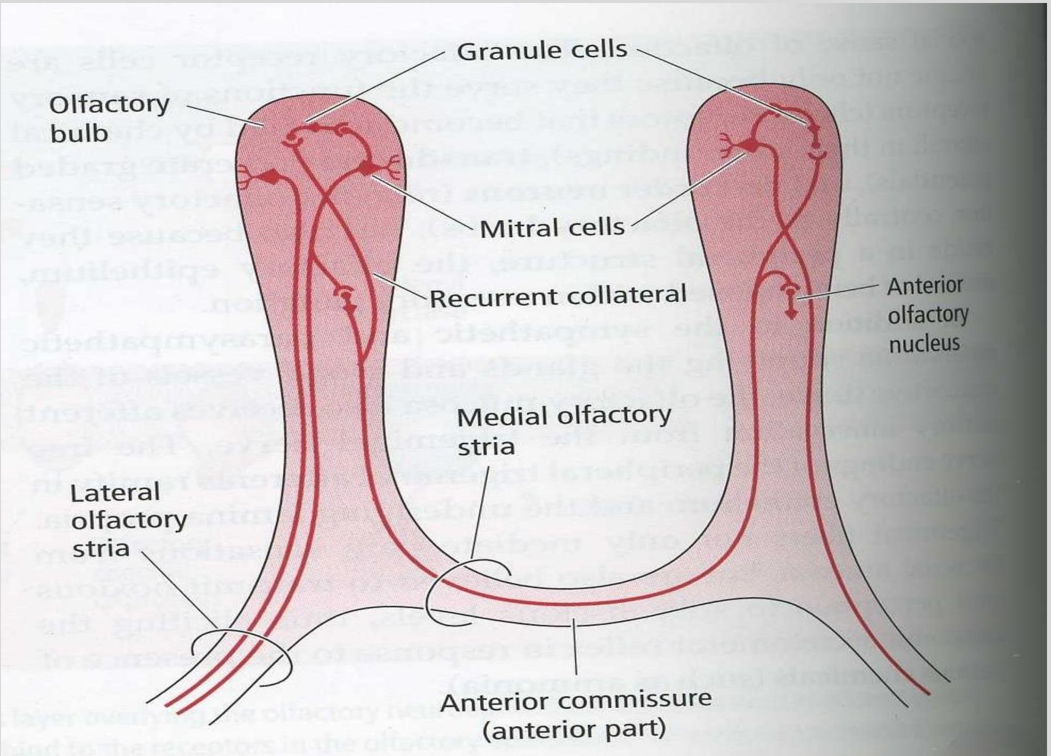
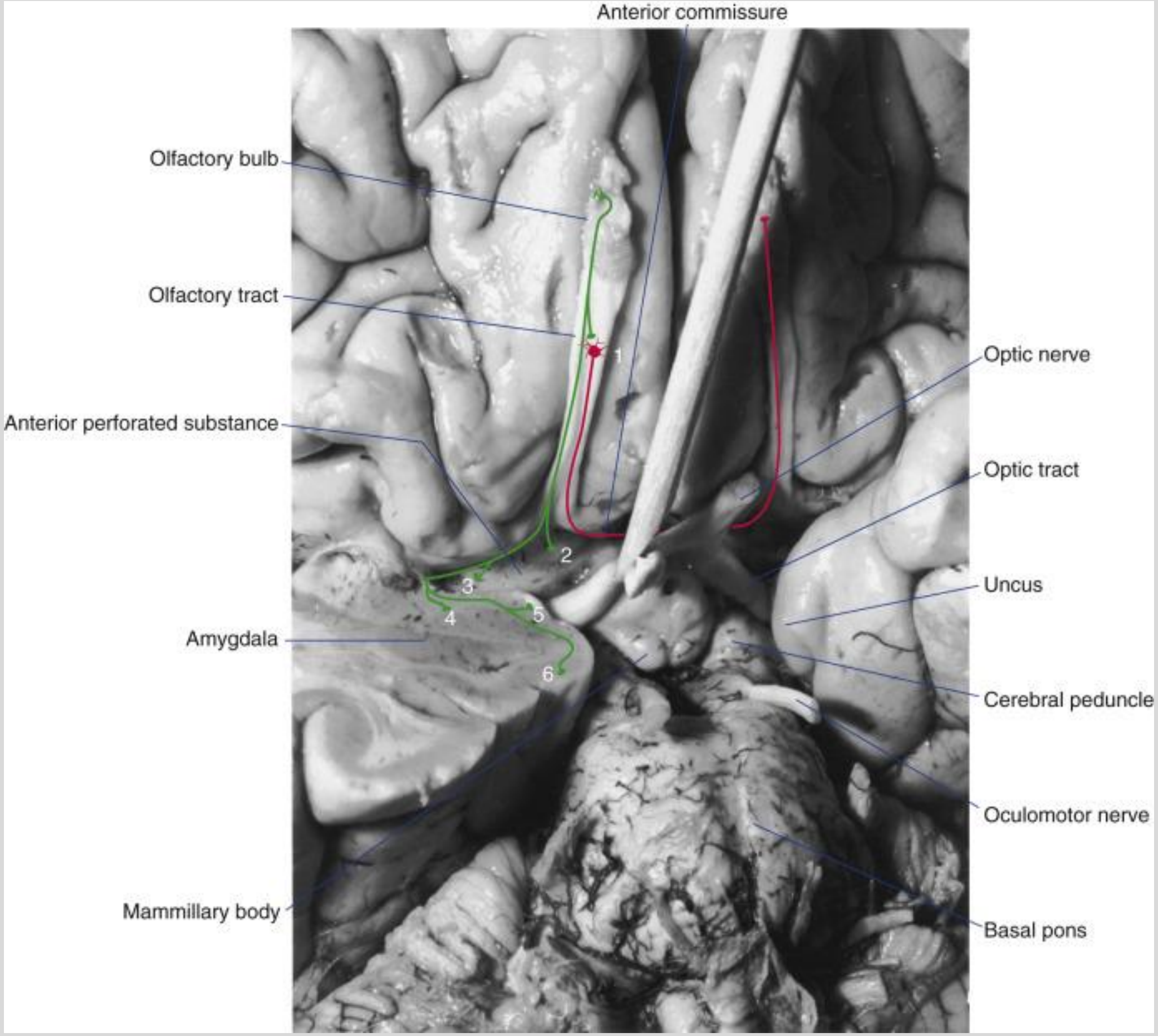


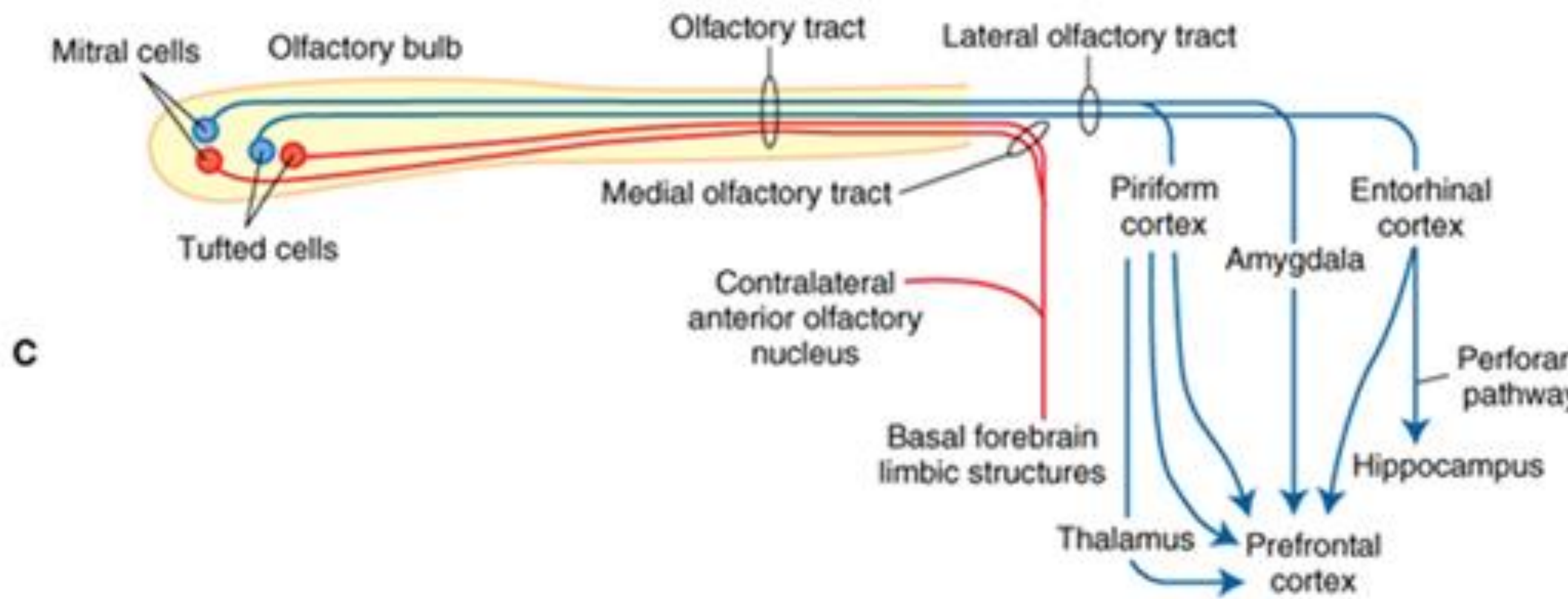
Figure 19.4 ● The olfactory bulb, anterior olfactory nucleus, and olfactory tract, and its division into the medial and lateral olfactory striae.

Olfactory Tracts



1. anterior olfactory nucleus
2. olfactory tubercle
3. piriform cortex
4. part of the amygdala
5. periamygdaloid cortex
6. parahippocampal gyrus, part of the entorhinal cortex

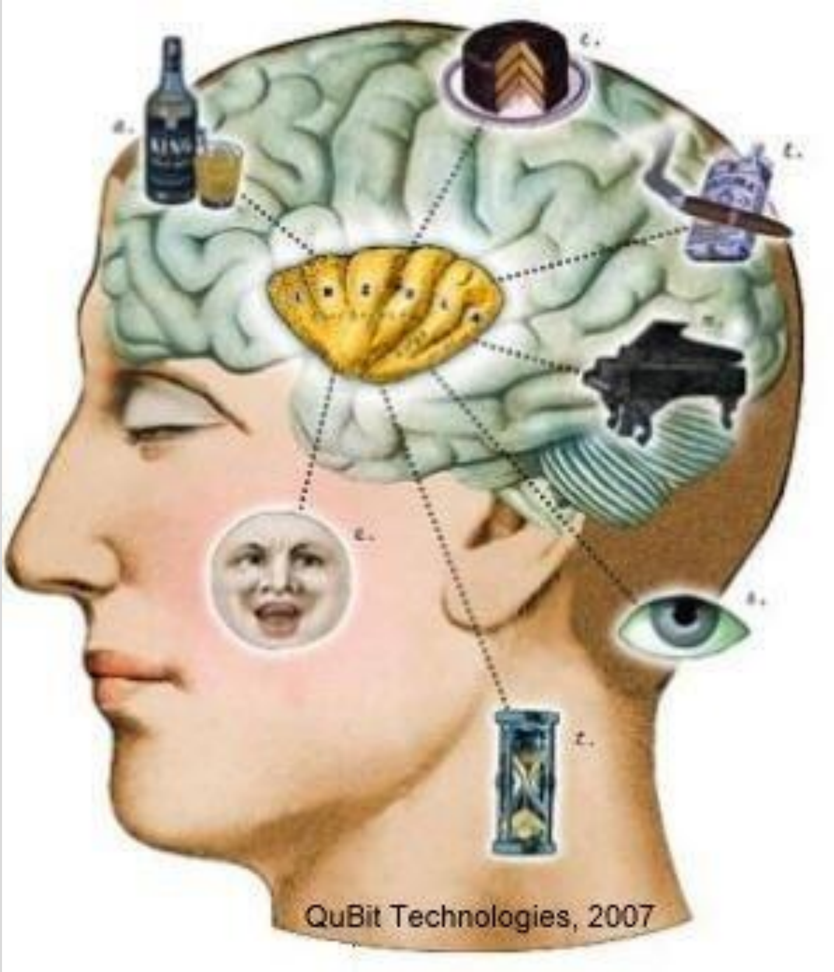
Olfactory bulb projections



- Anterior olfactory nucleus
  - Connects the two olfactory bulbs through a portion of the anterior commissure
- Piriform cortex (Part of the **Limbic System**)
  - Entorhinal area (Brodmann's area 28)
  - Dorsal entorhinal area (Brodmann's area 34)
  - Amygdala

Mediodorsal Thalamic Nucleus

Receives projections from
piriform cortex:"olfactory" thalamus



Projects to:

- Insular cortex
  - **Integration of smell & taste**
- Orbitofrontal cortex
  - **Reward & motivation**

Glomerular representation of odors is lost in piriform cortex

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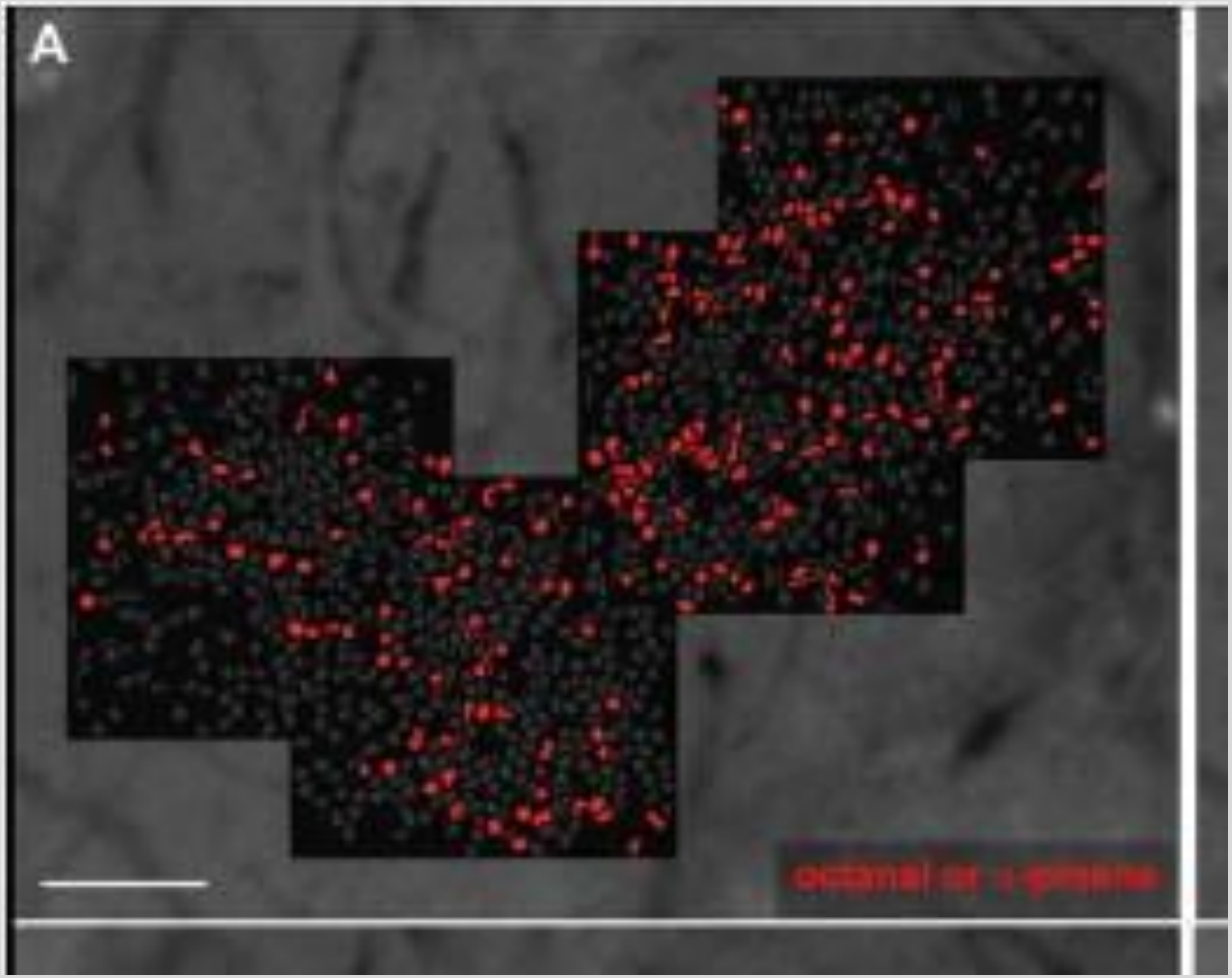

Stettler and Axel, Neuron, 2009

Olfactory Disorders: quantitative olfactory disorders

Normosmic: Normal smell sensation

Hyposmic: Decreased smell sensation

Anosmic: Absence of smell sensation

Sensitivity

Olfactory Disorders

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Anosmia: Absence of smell sensation

1) Recent nationally representative data reveals that anosmia afflicts 3.2% of US adults who are aged more than 40 years (3.4 million people) and this number increases with age (14–22% of those 60 years and older; Kern et al. 2014; Pinto et al. 2014; Hoffman et al. 2016).
2) The most common causes are sinonasal diseases, postinfectious disorder, and post-traumatic disorder.
3) Other etiologies (e.g., congenital, idiopathic, toxic disorders, or disorders caused by a neurodegenerative disease) are less common

Chem Senses. 2017 Anosmia—A Clinical Review

Qualitative Olfactory Disorders

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1) Parosmia: Stimulus elicits false sensation
   - smelling burnt paper instead of baby powder
   - smelling gasoline instead of coffee
   - Typically, the vast majority of patients describes these parosmic experiences as unpleasant
   - Parosmia is associated with loss of olfactory function
   - Parosmia might present after COVID 19 induced anosmia(7.1% of patients)
   - Cacosmia term is used when the perceived smell is unpleasant

2) Phantosmia: Random sensation of smell in the absence of stimulus

- Phantosmia is NOT associated with loss of olfactory function

- Distortion of olfactory perception: diagnosis and treatment, Chemical Senses 2002

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Clinical Correlations -Syndromes

- Olfactory dysfunction is a hallmark of certain syndromes
  - **Kallman Syndrome**
    - Hypogonadotropic hypogonadism (lack of pituitary hormones: LH & FSH→delayed puberty)
    - Anosmia
  - **Foster Kennedy Syndrome**
    - Meningioma of the olfactory groove compresses the olfactory tract and optic nerve
    - Ipsilateral anosmia, optic nerve atrophy and possibly contralateral papilledema (tumor mass effect)

Clinical Correlations - Age Related Disease

- Deterioration of the sense of smell occurs with normal aging
- Early symptom of Neurodegenerative disorders like **Parkinson's and Alzheimer's diseases**

Clinical Correlations – Trauma & Seizures

Trauma: Fracture of the floor of the anterior cranial fossa

- Often involve the cribiform plate
- **Anosmia and CSF rhinorrhea**

Uncinate fits: Aura of foul-smelling odors occur before seizure onset

- **Partial complex epilepsy** with a medial temporal focus (uncus)

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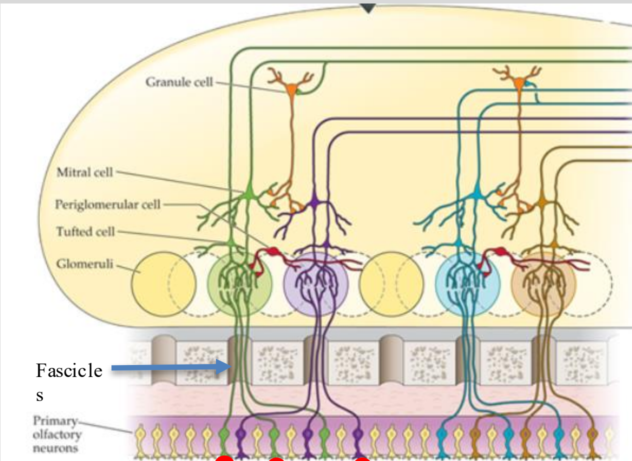
Allergic Rhinitis

- Cause of bilateral hyposmia
- Biopsies of the olfactory mucosa have shown that the sensory epithelial cells are still present, BUT their cilia are deformed and shortened and are buried under other mucosal cells

Korsakoff Psychosis and the Medial Thalamic Nuclei

- Alcoholics with Korsakoff psychosis also have a defect in odor discrimination
- Primarily caused by a deficiency in Thiamine (Vit B1)
- Symptoms include decline in cognitive function and confabulation
- In this disorder, anosmia is presumably caused by degeneration of neurons in the higher-order olfactory systems involving the medial thalamic nuclei.
  - Medial thalamic nuclei is a category including the mediodorsal thalamic nucleus

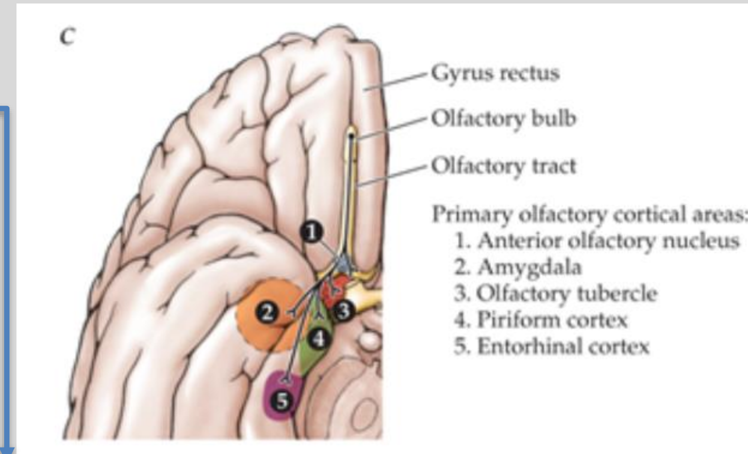
Overview of the Olfactory Pathway



5. PON synapse on second-order neurons (mitral and tufted cells)

4. Small fascicles converge onto glomeruli in the olfactory bulb

3. Axons of the PON collect into small fascicles which pass through the cribriform plate

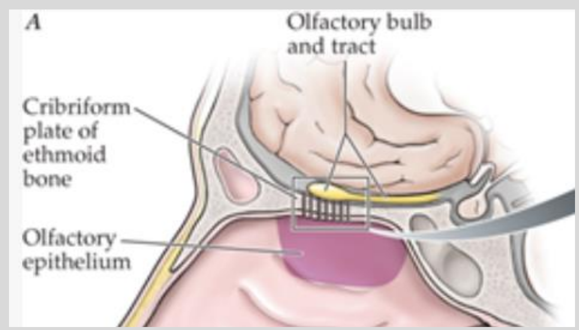


2. Odorants bind to the GPCR primary olfactory neurons (PON) in the olfactory epithelium

6. Second order neurons pass through the olfactory tract

7. Olfactory tract then provides input to the primary olfactory cortical areas

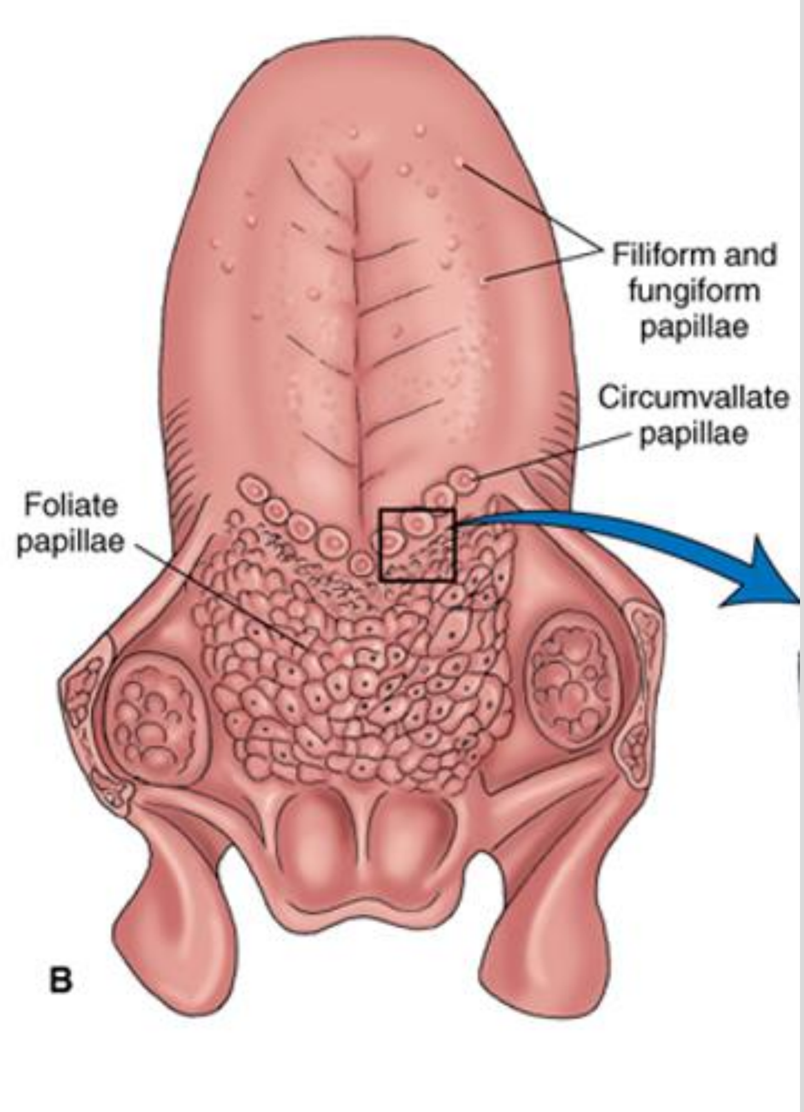
1. Odorants in the environment travel via ortho- or retro-nasal olfaction



[Chemical Senses: Taste and Smell | Neuroanatomy: Text and Atlas, 5e | AccessMedicine | McGraw Hill Medical]

Taste

Taste receptors in the tongue



Source: Siegel , Essential Neuroscience

Five basic sensations of taste: sweet, bitter, salty, sour, and umami. Umami is a Japanese word for pleasant savory taste. Glutamates and nucleotides umami taste sensation.

The sensation of taste are located in taste buds, which are the sensory organs for the taste system.

Taste buds contain the gustatory receptors

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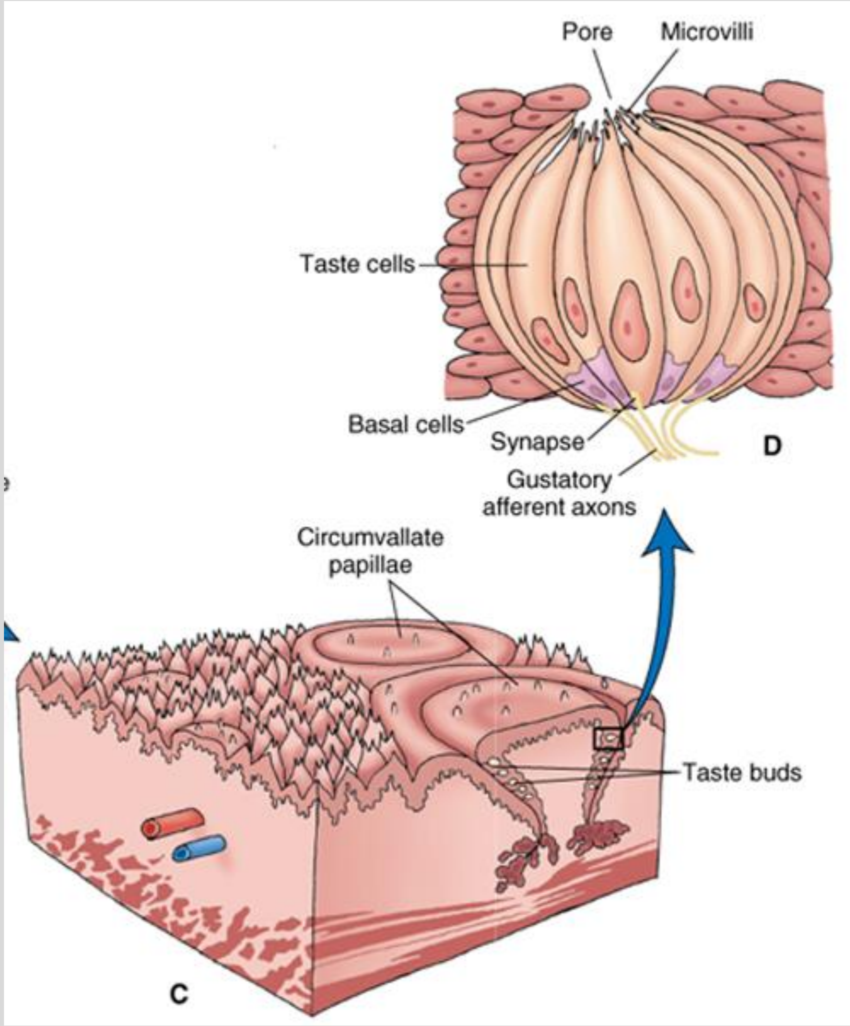
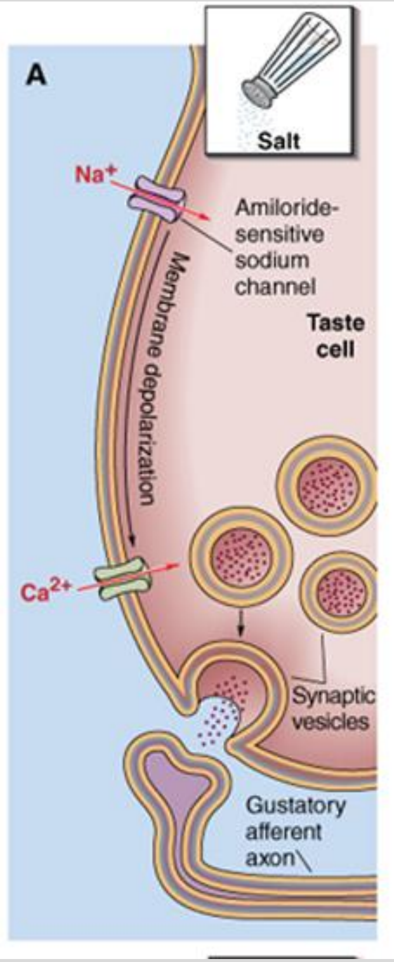


Source: Siegel , Essential Neuroscience

Taste buds are located in different types of papillae, which are protrusions on the surface of the tongue.

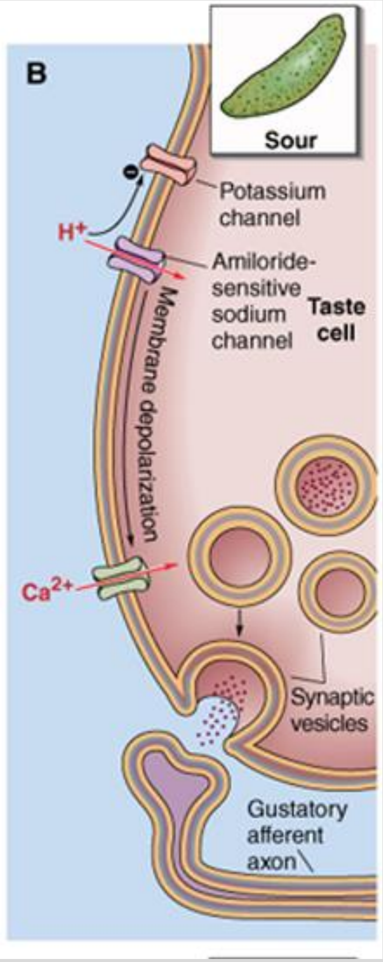
filiform, *fungiform*, *foliate*, and *circumvallate papillae*. The filiform and fungiform papillae are scattered throughout the surface of the anterior two thirds of the tongue, especially along the lateral margins and the tip.

Different tastes have different mechanisms of sensory transduction



- The transduction of salt taste is mediated by the influx of Na^+ (sodium) through the amiloride-sensitive Na^+ channels.
- Depolarization of the membrane opens voltage-activated calcium channels.
- Calcium triggers release of synaptic vesicles, activating gustatory afferent axons.

Different tastes have different mechanisms of sensory transduction



- Sour taste receptors detect protons H^+. and is mediated by closure of voltage-dependent K^+ (potassium) channels.
- Depolarization of the membrane opens voltage-activated calcium channels.
- Calcium triggers release of synaptic vesicles, activating gustatory afferent axons.

Different tastes have different mechanisms of sensory transduction

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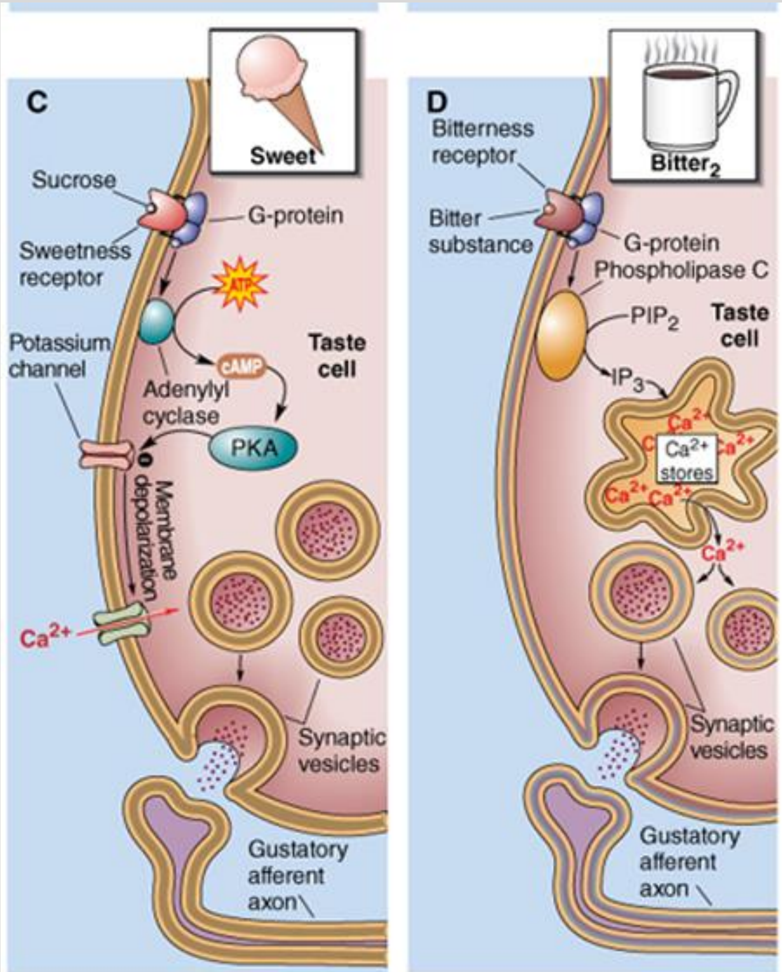

-Sweet and bitter taste use G protein coupled receptors.

-Sweet taste is mediated by activation of G proteins and adenylate cyclase, which increases the levels of cyclic adenosine monophosphate (cAMP), producing membrane depolarization.

-Bitter substances activate a G protein that activates phospholipase C and generates inositol triphosphate (IP_3) and producing calcium release from internal stores.

-Very different chemicals can taste bitter and are potentially toxic. There are more than 30 different bitter receptors.

Taste is transmitted by CN VII, IX, X

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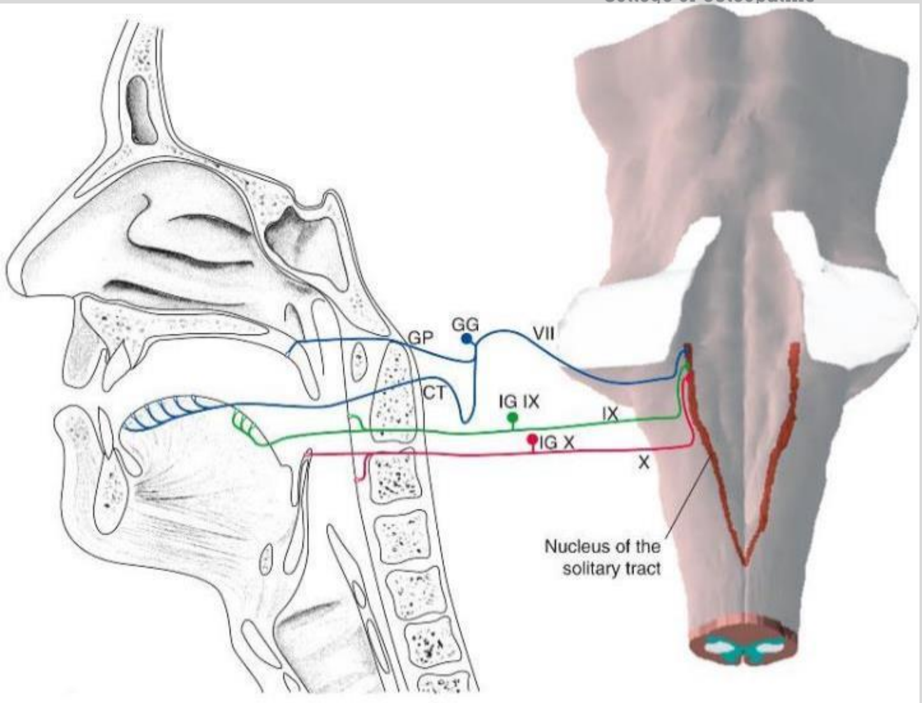
- The taste buds on the anterior two thirds of the tongue are innervated by the facial nerve (CN VII).
- Taste buds on the posterior one third of the tongue are innervated by the glossopharyngeal nerve (CN IX).
- Taste buds on the epiglottis and pharyngeal walls are innervated by the vagus nerve (CN X)
- The central processes of all three terminate in rostral parts of the nucleus of the solitary tract in the dorsal medulla, close to the 4th ventricle.



Nolte's The Human Brain Eighth Edition

Taste Pathways

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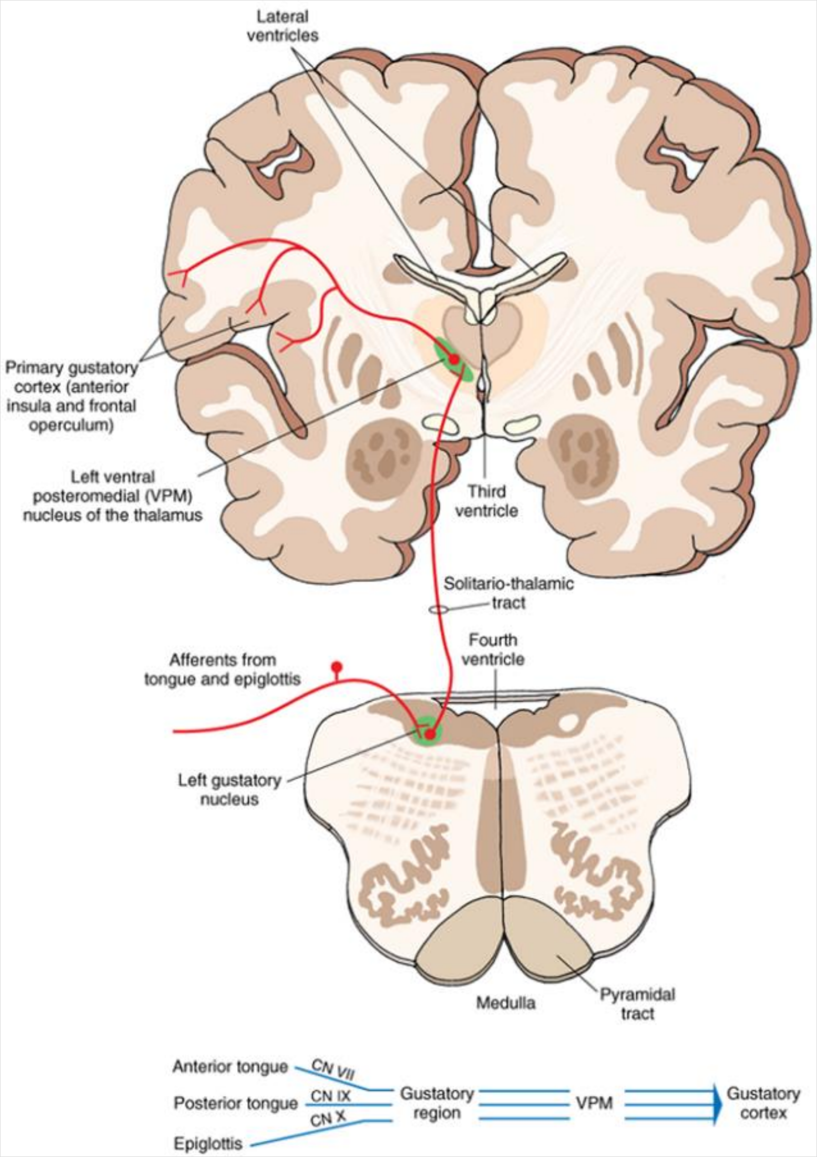

Nolte's The Human Brain Eighth Edition

Solitary tract nucleus cells respond to multiple tastes compared to gustatory cortex

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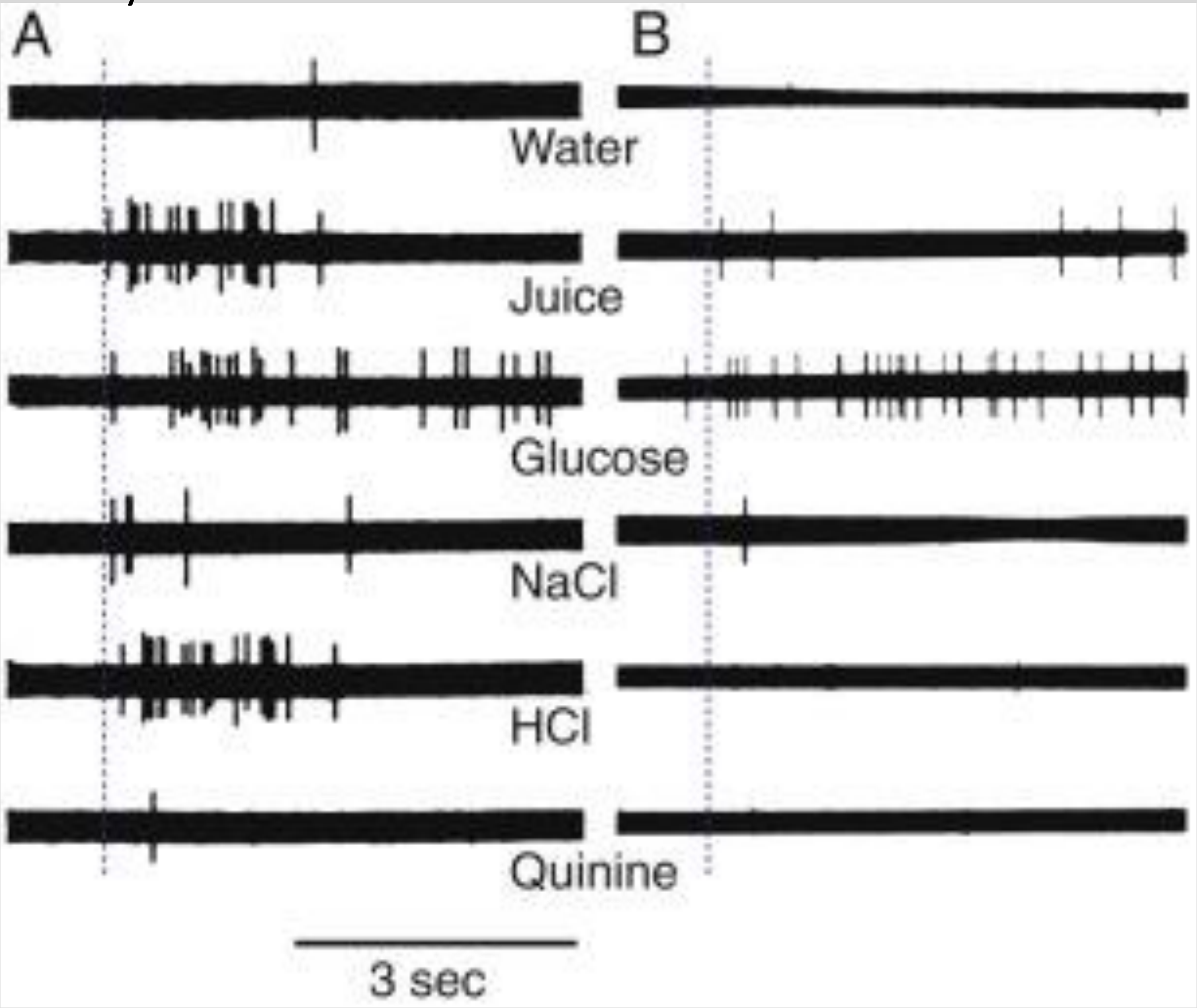
Solitary tract nucleus cells

Gustatory cortex



Conditions affecting gustatory system (Ageusia)

Heavy smoking

- Determination with tasting solutions has showed that smokers' detection thresholds were significantly elevated for salt, acid, sucrose or quinine

Bell's palsy

- The taste buds on the anterior two thirds of the tongue are innervated by the facial nerve (CN VII).

Chemo-/radiotherapy in the head/neck

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- # Summary Slide

The olfactory nerve (CNI) consists of sensory neuron axons that synapse in the olfactory bulb. CNI carries only one single modality: Special Visceral Afferent.

There is no incoming precortical relay of olfactory information in the thalamus. Then, olfactory information travels via the mediodorsal nucleus of the thalamus to the insula followed by the orbitofrontal cortex.

Neurodegenerative diseases, such as Alzheimer’s and Parkinson’s disease, are often accompanied by hyposmia.

Each taste bud harbors all five varieties of non-neuronal sensory cells that are connected to either CNVII, IX or X.

- Question #1

Neurodegeneration as seen in Alzheimer’s and Parkinson’s disease often causes a noticeable decline in odor sensitivity. This phenomenon is best described as:

A. Anosmia
B. Cacosmia
C. Hyposmia
D. Parosmia
E. Phantosmia

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- Question #2

A 15-year-old girl with a family history of benign meningiomas presents with left-sided absence of smell sensation and contralateral papilledema. She points out that her symptoms have worsened over time. Her most likely diagnosis is:

A. Cacosmia and uncinate epileptic seizure
B. Cribriform plate fracture
C. Early-onset Parkinson’s disease
D. Foster Kennedy syndrome
E. Kallmann syndrome

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- Question #3

A 15-year-old girl who has been unable to smell any odors since birth presents to her pediatrician with the Complaint that she is not developing any secondary sex characteristics, which would have been otherwise expected for her age. Her diagnosis is consistent with:

A. Congenital compression of the olfactory groove
B. Early-onset Alzheimer's disease
C. Foster Kennedy syndrome
D. Kallmann syndrome
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Source: http://what-when-how.com/wp-content/uploads/2012/04/tmp3662_thumb222.jpg

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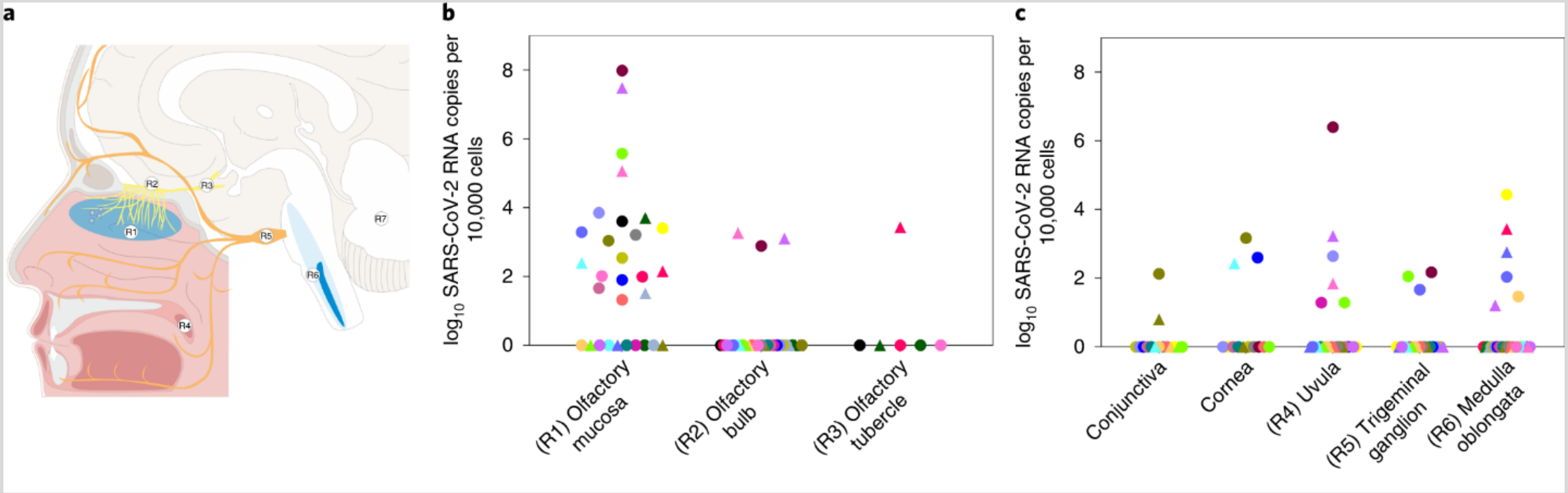
Source: http://what-when-how.com/wp-content/uploads/2012/04/tmp3662_thumb222.jpg

- Tropism of SARS-CoV-2



Olfactory transmucosal SARS-CoV-2 invasion as a port of central nervous system entry in individuals with COVID-19

Meinhardt et al, Nature neuroscience

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Acknowledgments

Kassandra Lee Sturm, OMS III.

Feedback on lecture:

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